

LAB SHEET: WEB DESIGN TECHNOLOGY (LAB 3.2)

TITLE: CONSTRUCTING AN INTERACTIVE STUDENT PORTFOLIO WITH BOOTSTRAP 5

TOPIC: 3.2 Construct Interactive Web Elements with Bootstrap

STUDENT INFORMATION

Name	Registration No	Date
NUR IRDINA WAFIYYA BINTI NIK FAZZELLY	10DIT25F1128	10/03/2026

1. LEARNING OUTCOMES

By the end of this session, you will be able to:

- **Implement** professional Bootstrap Modals for data entry (3.2.1).
- **Enhance** User Experience (UX) using interactive Tooltips and Popovers (3.2.2).
- **Construct** an Advanced Carousel featuring a custom progress bar and thumbnail navigation (3.2.3).

2. THEORETICAL OVERVIEW

To create a high-fidelity interface, we will combine Bootstrap 5 with Custom JavaScript:

- **The Modal:** A "child" window that pops up to focus user attention.
- **Popper.js Components:** Tooltips (hover) and Popovers (click). *Note: These are not auto-enabled; we must "wake them up" using JavaScript!*
- **Carousel Events:** We use the `slid.bs.carousel` event to sync our progress bar and thumbnails in real-time.

3. STEP-BY-STEP IMPLEMENTATION

Step 1: Document Setup & Design Language

Create a file named `portfolio.html`. We'll define a "Primary Purple" theme to make your portfolio look modern and premium.

Action: Setup the <head> and define the CSS variables:

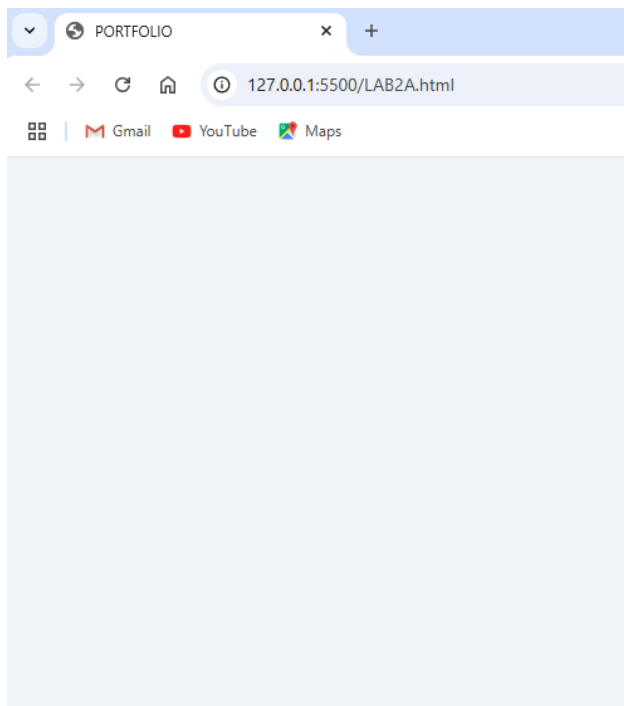
HTML

```
<style>
  :root {
    --primary-purple: #5d46b8;
    --bg-soft: #f0f4f8;
  }
  body { background-color: var(--bg-soft); font-family: 'Segoe UI',
sans-serif; }
  .navbar-custom { background-color: var(--primary-purple); }
  .card { border: none; border-radius: 12px; box-shadow: 0 4px 10px
rgba(0, 0, 0, 0.05); margin-bottom: 20px; }

  /* Advanced Carousel Styling */
  .carousel-demo img { height: 400px; object-fit: cover; border-
radius: 8px; }
  .active-thumb { border: 3px solid var(--primary-purple)
!important; opacity: 1 !important; }
  .thumbs img { width: 100px; height: 60px; object-fit: cover;
cursor: pointer; opacity: 0.6; transition: 0.3s; }

  #btnTop { display: none; position: fixed; bottom: 30px; right:
30px; z-index: 99;
    background-color: var(--primary-purple); color: white;
border: none;
    padding: 15px; border-radius: 50%; transition: 0.3s; }
</style>
```

 **Checkpoint 1:** Screenshot your blank page/setup and explain why we use CSS :root variables.



We use CSS `:root` variables to define global, reusable custom properties that cascade throughout the entire document, which significantly enhances code maintainability, enables powerful dynamic styling, and improves readability.

Step 2: Profile Section (Tooltips & Popovers)

Let's add your identity! We will use **Tooltips** for skill badges and a **Popover** for your bio.

Action: Insert the Navbar and Profile Card into the <body>:

HTML

```
<nav class="navbar navbar-expand-lg navbar-dark navbar-custom mb-4">
  <div class="container justify-content-center">
    <ul class="navbar-nav">
      <li class="nav-item"><a class="nav-link active"
href="#"><i class="fas fa-user-circle"></i> My Profile</a></li>
      <li class="nav-item"><a class="nav-link" href="#"><i
class="fas fa-laptop-code"></i> Web Tech</a></li>
    </ul>
  </div>
</nav>

<div class="container pb-5">
  <div class="row">
    <div class="col-lg-4">
      <div class="card p-3 text-center">
        
        <h4 class="fw-bold mb-1">Your Name</h4>
        <p class="text-muted small">ID: 10DITXXXXXX</p>
        <div class="d-flex flex-wrap justify-content-center
gap-2 mt-2">
          <span class="badge bg-primary" data-bs-
toggle="tooltip" title="Final Year Student">Malaysia</span>
          <span class="badge bg-secondary" data-bs-
toggle="tooltip" title="HTML/CSS/JS">Frontend Dev</span>
        </div>
      </div>
      <div class="card p-3">
        <h5><i class="fas fa-info-circle text-primary me-
2"></i>Quick Bio</h5>
        <button type="button" class="btn btn-sm btn-outline-
primary" data-bs-toggle="popover"
          data-bs-trigger="focus" title="Academic
Background"
          data-bs-content="Currently pursuing Diploma
in IT.">View Detailed Bio</button>
      </div>
    </div>
  </div>
</div>
```

 **Checkpoint 2:** Show your Profile Section. Hover your mouse over the "Malaysia" badge to show the Tooltip.

Lab 3.1.1: JavaScript Fundamentals

Variables, Data Types and Operators

My Profile Web Tech



Nur Iridna Wafiyya

ID: 10DIT25F1128

Malaysia

Frontend Dev

Quick Bio

[View Detailed Bio](#)

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Step 3: The Advanced Project Carousel

Time to show off your work! This carousel includes a **Progress Bar** and **Thumbnails**.

Action: Add this inside the `col-lg-8` div:

HTML

```
<div class="col-lg-8">
  <div class="card shadow-sm border-0">
    <div class="card-body p-4">
      <header class="mb-4 text-center">
        <h2 class="h4 fw-bold text-
primary">Project Showcase</h2>
        <p class="text-muted small">Hover to
pause, click thumbnails to navigate.</p>
      </header>

      <div id="demoCarousel" class="carousel slide
carousel-demo" data-bs-ride="carousel">
        <div class="carousel-inner shadow-sm
rounded">
          <div class="carousel-item active">
            
            <div class="carousel-caption d-
none d-md-block bg-dark bg-opacity-50 rounded">
              <h5>User Interface
Design</h5>
              <p>Crafting modern layouts
with Bootstrap 5 Grid and Utility classes.</p>
            </div>
```

</div>
<pre> <div class="carousel-item"> <div class="carousel-caption d-none d-md-block bg-dark bg-opacity-50 rounded"> <h5>Database Integration</h5> <p>Dynamic data management with PHP & MySQL structural logic.</p> </div> </div> <div class="carousel-item"> <div class="carousel-caption d-none d-md-block bg-dark bg-opacity-50 rounded"> <h5>JavaScript Interactivity</h5> <p>Mastering DOM manipulation and Event Listeners for seamless UX.</p> </div> </div> </div> </pre>
<pre> <button class="carousel-control-prev" type="button" data-bs-target="#demoCarousel" data-bs-slide="prev"> </button> <button class="carousel-control-next" type="button" data-bs-target="#demoCarousel" data-bs-slide="next"> </button> </div> <div class="progress mt-3" style="height:6px;"> <div id="carouselProgress" class="progress-bar progress-bar-striped progress-bar-animated bg-info" style="width:0%"></div> </div> <div class="thumbs d-flex justify-content-center gap-2 mt-4"> </pre>

```


</div>
</div>
</div>
</div>
<div class="card p-4 border-start border-info border-5 shadow-sm">
  <div class="d-flex justify-content-between align-items-center">
    <div>
      <h5 class="fw-bold mb-1">Class
Registration</h5>
      <p class="text-muted small mb-0">Join the Web
Design Technology session for Semester 2.</p>
    </div>
    <button type="button" class="btn btn-info text-
white fw-bold px-4" data-bs-toggle="modal"
      data-bs-target="#registrationModal">
      <i class="fas fa-user-plus me-2"></i>Register
    </button>
    </div>
  </div>
</div>
</div>
</div>
</div>

```

Step 4: Registration Modal & Footer

We will add a "Call to Action" button that opens a beautiful modal.

Action: Add the Modal at the very bottom of your <body>:

HTML

```

<div class="modal fade" id="registrationModal" tabindex="-1">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content border-0">
      <div class="modal-header bg-info text-white">
        <h5 class="modal-title">Workshop Enrollment</h5>
        <button type="button" class="btn-close btn-close-
white" data-bs-dismiss="modal"></button>
      </div>
      <form id="regForm">
        <div class="modal-body p-4">
          <label class="form-label fw-bold">Full
Name</label>
          <input type="text" class="form-control mb-3"
id="formName" required>
          <label class="form-label fw-bold">Student
ID</label>
          <input type="text" class="form-control"
placeholder="DITXXXX" required>
        </div>
        <div class="modal-footer">
          <button type="submit" class="btn btn-info text-
white px-4">Submit Application</button>
        </div>
      </form>
    </div>
  </div>
</div>

```

```

        </form>
      </div>
    </div>
  </div>

  <button onclick="topFunction()" id="btnTop" title="Go to top"><i class="fas fa-arrow-up"></i></button>

  <footer class="navbar-custom py-4 mt-5">
    <div class="container text-center text-white">
      <p class="mb-1">© Copyright 2026 - JTMK Polytechnic</p>
      <p class="mb-0" style="font-size: 0.85rem; opacity: 0.8;">Prepared by: Shu Haila Mohd Yusof (Reg. ID.: 10DIT26F1705)</p>
    </div>
  </footer>

```

Step 5: Bringing It to Life (JavaScript)

This is where the magic happens! We will initialize all components.

Action: Add the script before the `</body>` tag:

JavaScript

```

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

<script>
  (function () {
    // Initialize Tooltips & Popovers (3.2.2)
    const tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="tooltip"]');
    tooltipTriggerList.forEach(el => new bootstrap.Tooltip(el));

    const popoverTriggerList = document.querySelectorAll('[data-bs-toggle="popover"]');
    popoverTriggerList.forEach(el => new bootstrap.Popover(el));

    // Advanced Carousel Logic (3.2.3)
    const carouselEl = document.querySelector('#demoCarousel');
    const intervalMs = 4000;
    const bar = document.getElementById('carouselProgress');
    const thumbs = document.querySelectorAll('.thumbs img');
    const carousel = new bootstrap.Carousel(carouselEl, { interval: intervalMs,
    'hover' });

    let progressTimer;
    function startProgress() {
      clearInterval(progressTimer);
      bar.style.width = '0%';
      const start = Date.now();
      progressTimer = setInterval(() => {
        const elapsed = Date.now() - start;
        const pct = Math.min(100, (elapsed / intervalMs) * 100);
        bar.style.width = pct + '%';
        if (pct >= 100) clearInterval(progressTimer);
      }, intervalMs);
    }
  })();

```

```

        }, 50);
    }

    // Sync Thumbnails and Progress Bar
    carouselEl.addEventListener('slid.bs.carousel', (e) => {
        thumbs.forEach(t => t.classList.remove('active-thumb'));
        thumbs[e.to].classList.add('active-thumb');
        startProgress();
    });

    // Thumbnail Navigation
    thumbs.forEach((thumb, idx) => {
        thumb.addEventListener('click', () => carousel.to(idx));
    });

    carouselEl.addEventListener('slide.bs.carousel', () => clearInterval(progressInterval));

    startProgress(); // Start progress for the first slide

    // Back to Top logic
    let mybutton = document.getElementById("btnTop");
    window.onscroll = () => { mybutton.style.display = (window.scrollY > 100) ?
"none"; };

    // Registration Feedback
    document.getElementById('regForm').addEventListener('submit', function (e) {
        e.preventDefault();
        alert("Success! Registration for " + document.getElementById('formName').value +
" is complete.");
        bootstrap.Modal.getInstance(document.getElementById('registrationModal')).hide();
    });

    function topFunction() {
        window.scrollTo({ top: 0, behavior: 'smooth' });
    }
</script>

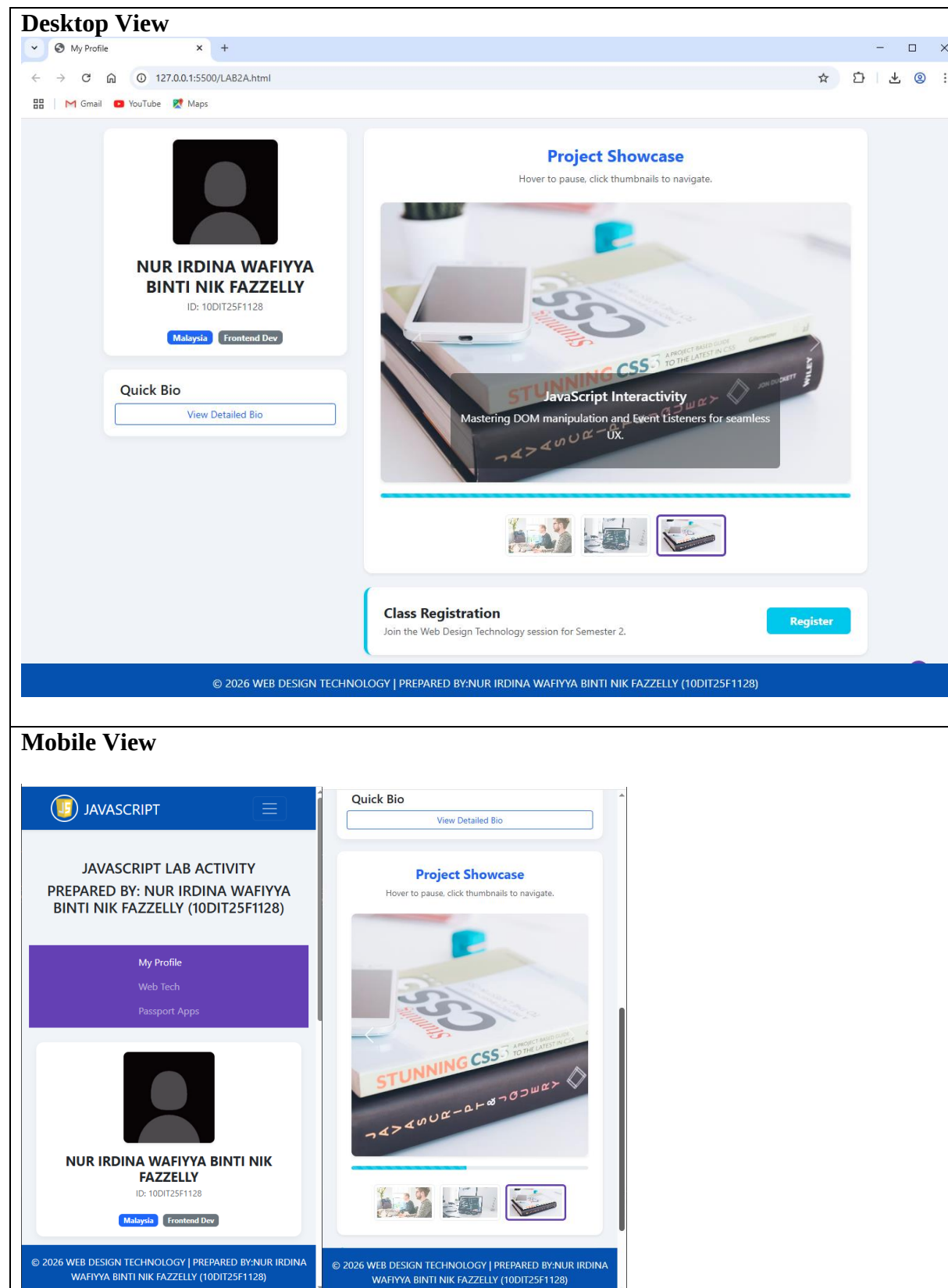
```

4. STEP-BY-STEP VALIDATION:

Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1: Setup	(Insert Image)	Explain the CSS variables/layout...
Step 2: Profile	(Insert Image)	Explain how Tooltips/Popovers work...
Cont...		

5. SCREENSHOT OUTPUT (COMPLETE):



6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations:
`Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands! ✨